REPUBLIC OF THE PHILIPPINES
MINDANAO STATE UNIVERSITY – ILIGAN INSTITUTE OF TECHNOLOGY
INTEGRATED DEVELOPMENTAL SCHOOL
JUNIOR & SENIOR HIGH SCHOOL DEPARTMENT • ILIGAN CITY, LANAO DEL NORTE

COURSE CURRICULAR ASSESSMENT DOCUMENT

TABLE OF SPECIFICATIONS (TOS) & COMPLETE QUESTION BANK

Course Worksheets (Quizzes 1 to 5) and 1st Quarter Examination

Technical-Vocational-Livelihood (TVL) / ICT — Computer Systems Servicing & Electronics

Course Title & Code	**Information Technology 10 (IT 10)**	**Academic Term**	**First Quarter, S.Y. 2026–2027**
Subject Teacher	**Alan L. Vergara**	**Grade Level**	**Grade 10 (Junior High School)**
Assessment Delivery	**MIOW LMS (Online Testing & Virtual Arduino Lab)**	**Curriculum Track**	**DepEd K to 12 TLE-ICT Electronics & Microcontrollers**
Exam Bank Items	**77 Items (50 drawn per attempt; 12 required)**	**Worksheet Quizzes**	**5 Worksheets (24–26 items each, 122 total items)**
Total Question Bank	**199 Authored Assessment Items with Keys & Rubrics**	**Cognitive Model**	**Bloom's Revised Taxonomy (DepEd Standards)**

Executive Summary & Assessment Architecture

This official curriculum assessment repository establishes the pedagogical alignment between instructional competencies, cognitive skill domains, and evaluation instruments for **Information Technology 10** at the MSU-IIT Integrated Developmental School. It encompasses both formative classroom worksheets (quizzes) and the summative 1st Quarter Examination.

- **Formative Worksheets (Quizzes 1–5):** 10-minute timed assessments featuring randomized draws, SVG resistor diagrams, and rubric-scored Virtual Arduino Lab simulations.
- **Summative 1st Quarter Examination:** 45-minute comprehensive exam utilizing a 77-item bank. Each student receives 50 items (12 mandatory items: 10 visual component identifications + 2 laboratory problem-solving tasks, plus 38 unseen-first items).
- **DepEd Alignment:** Structured across Remembering (Recall), Understanding (Comprehension), Applying (Computation/Decoding), and Analyzing/Creating (Circuit Design & Code Synthesis).

ALAN L. VERGARA	**DEPARTMENT HEAD / COORDINATOR**	**SCHOOL PRINCIPAL / DIRECTOR**
Subject Teacher / Course Author	Science & Technology Department, IDS	Integrated Developmental School, MSU-IIT

SECTION 1: MASTER COURSE TABLE OF SPECIFICATIONS (TOS)

Master cognitive blueprint aggregating all instructional units, learning hours, item distributions, and percentage weights for IT 10 (First Quarter).

Unit / Curricular Competency	Teaching Hours	Weight (%)	Total Items	Cognitive Process Dimensions (Bloom's Revised Taxonomy)				Target Assessments / Delivery Mode
				Remembering	Understanding	Applying	Analyzing / Creating	
Unit 1: The Arduino Architecture & Hardware ATmega328P microcontroller, digital/analog pins, power headers, USB, reset, bare minimum sketch structure.	10	25.0%	48	16	22	10	0	Quiz — Arduino Part 1 Quiz — Arduino Part 2 Exam Domain A & D
Unit 2: Prototyping & Laboratory Wiring Solderless breadboards, jumper wires, LED polarity, current-limiting series resistors, safety precautions.	6	15.0%	24	8	10	6	0	Quiz — Lab Activity 1 Exam Domain D
Unit 3: Resistor Color Codes & Ohm's Law 4-band and 5-band color charts, multipliers, gold/silver tolerances, Ohm's law (R = V/I) computations.	8	20.0%	44	11	13	20	0	Quiz — Resistor Color Code Exam Domain B & C
Unit 4: Arduino C/C++ Programming & I/O IDE setup, pinMode(), digitalWrite(), analogRead(), Serial Monitor baud rates, variables (int/float).	8	20.0%	39	10	16	13	0	Quiz — Programming with Arduino Exam Domain A
Unit 5: Visual Component Identification Physical identification of microcontrollers, breadboards, sensors, actuators, and hardware modules.	4	10.0%	20	15	5	0	0	Exam Part I (Required visual photos)
Unit 6: Virtual Lab Simulation & Synthesis Hands-on circuit construction on simulation canvas, code development, multi-LED traffic light, button I/O.	4	10.0%	24	2	2	6	14	Quiz Part 2 Lab Tasks Exam Part IV Problem Solving
TOTAL CURRICULAR BANK	**40 hrs**	**100.0%**	**199**	**62 (31.2%)**	**68 (34.2%)**	**55 (27.6%)**	**14 (7.0%)**	**All 5 Quizzes + 1st Quarter Exam**

Assessment Administration & Test Security Policy

Under the approved IT 10 assessment guidelines (SY 2026–2027), all examinations and quizzes adhere to the following execution protocols:

- **Fixed Window Gating:** Assessments open strictly within scheduled timeframes (e.g., Friday 08:00–10:00 Asia/Manila). Submissions outside the window are server-blocked.
- **Randomized Unseen-First Draw:** Each quiz draws a fresh 20-item subset from its authored bank, while the exam draws 50 items (12 mandatory + 38 randomized), guaranteeing integrity.

- **Grading & Review Policy:** Immediate score visibility upon submission; answer keys and detailed explanations are withheld until official teacher release.

SECTION 2: INDIVIDUAL WORKSHEET & EXAMINATION TABLES OF SPECIFICATIONS

Detailed DepEd-format Table of Specifications for each individual course worksheet (quiz) and the 1st Quarter Examination.

Worksheet 1: Quiz — Arduino Part 1 (24 Items)

Topic / Competency	Items	%	Remembering	Understanding	Applying	Item Numbers
A. Arduino Uno Board & Microcontroller Architecture	12	50%	7	4	1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
B. Circuit Components & Beginner Boards (Nano/Mega, Breadboard, LED, Pushbutton)	12	50%	5	4	3	13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24
TOTAL	**24**	**100%**	**12 (50.0%)**	**8 (33.3%)**	**4 (16.7%)**	**Items 1–24**

Worksheet 2: Quiz — Lab Activity 1: Blinking LED (24 Items)

Topic / Competency	Items	%	Remembering	Understanding	Applying	Item Numbers
A. Circuit Assembly, Series Limiting Resistor & Hardware Safety	10	41.7%	4	4	2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
B. Arduino IDE Workflow, Blink Sketch Syntax & Timing Logic	14	58.3%	4	6	4	11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24
TOTAL	**24**	**100%**	**8 (33.3%)**	**10 (41.7%)**	**6 (25.0%)**	**Items 1–24**

Worksheet 3: Quiz — Arduino Part 2 (26 Items)

Topic / Competency	Items	%	Remember	Understand	Apply	Create	Item Numbers
A. Open-Source Prototyping, Bare Minimum & Libraries	9	34.6%	3	5	1	0	1, 2, 3, 4, 5, 6, 7, 8, 9
B. Sensors, Actuators & Single-Board Computers	11	42.3%	5	4	2	0	10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20
C. Traffic Light Assessment & Circuit Simulation	4	15.4%	1	1	2	0	21, 22, 23, 24
D. Virtual Arduino Lab Simulation Tasks (Traffic Light & Button LED)	2	7.7%	0	0	0	2	25, 26 (Required Tasks)
TOTAL	**26**	**100%**	**9 (34.6%)**	**10 (38.5%)**	**5 (19.2%)**	**2 (7.7%)**	**Items 1–26**

Worksheet 4: Quiz — Resistor Color Code (24 Items)

Topic / Competency	Items	%	Remembering	Understanding	Applying	Item Numbers
A. 4-Band Resistor Reading Rules, Multipliers & Tolerances	10	41.7%	5	3	2	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
B. Resistance Value Decoding (4-Band Calculations)	4	16.7%	0	1	3	11, 12, 13, 14
C. 5-Band Precision Resistors (Significant Digits & Tolerances)	5	20.8%	1	2	2	15, 16, 17, 18, 19
D. Resistor Color Principles & Multimeter Verification	5	20.8%	1	3	1	20, 21, 22, 23, 24
TOTAL	**24**	**100%**	**7 (29.2%)**	**9 (37.5%)**	**8 (33.3%)**	**Items 1–24**

Worksheet 5: Quiz — Programming with Arduino (24 Items)

Topic / Competency	Items	%	Remembering	Understanding	Applying	Item Numbers
A. Course Guide Structure, Short Circuits & Breadboard Safety	6	25.0%	3	2	1	1, 2, 3, 4, 5, 6
B. IDE Environment, Verification, Uploading & Serial Monitor	6	25.0%	2	3	1	7, 8, 9, 10, 11, 12
C. Digital & Analog I/O Functions (digitalWrite/Read, analogRead/Write)	6	25.0%	1	3	2	13, 14, 15, 16, 17, 18
D. Sensor Interfacing & C/C++ Data Types (int, float, LDR, TMP36)	6	25.0%	2	3	1	19, 20, 21, 22, 23, 24
TOTAL	**24**	**100%**	**8 (33.3%)**	**11 (45.8%)**	**5 (20.8%)**	**Items 1–24**

Summative Examination: 1st Quarter Examination (77 Items)

Official 77-item question bank blueprint. Each student attempt draws 50 items: 12 mandatory items (Part I visual identification + Part IV lab tasks) + 38 unseen-first items from Domains A–D.

Domain	Content / Learning Competency	Cognitive Skill	Bank Items	% Bank	Per Attempt	Item Number Range
Domain A	Arduino Programming Fundamentals (code constructs, setup/loop, pin modes, delay, I/O)	Application / Analysis	25	32.5%	~14 (of 38)	Items 1–15, 56–65
Domain B	Resistor Color Code (4-band decoding with band diagrams)	Application / Decoding	10	13.0%	~6 (of 38)	Items 16–25
Domain C	Resistance Value Computation (Ohm's Law: R = V/I)	Application / Computation	10	13.0%	~6 (of 38)	Items 26–35
Domain D	Arduino Family & SIK Components (Hardware Identification)	Knowledge / Recall	20	26.0%	~12 (of 38)	Items 36–55
Part I	Visual Component Identification (Real photograph matching)	Knowledge / Recall (Visual)	10	13.0%	**10 (Required)**	Items 66–75 (All attempts)
Part IV	Problem Solving in Virtual Arduino Lab (Circuit wiring + Sketch coding)	Analysis / Synthesis (Rubric)	2	2.6%	**2 (Required)**	Items 76–77 (All attempts)
TOTAL 1st QUARTER EXAMINATION		**Comprehensive**	**77**	**100.0%**	**50 Items Drawn**	**Items 1–77**

SECTION 3: COMPLETE QUESTION BANK WITH ANSWER KEYS & RUBRICS

Exhaustive question bank covering all 5 course worksheets and the 1st Quarter Examination. Includes full question stems, diagrams, answer options, keyed responses, accepted variants, and scoring rubrics.

Worksheet 1: Quiz — Arduino Part 1 (24 Items)

Item #1 Understanding 1 pt

Arduino is an _____ electronic prototyping platform.

A. open-source ✓ (Correct)
B. closed-source
C. paid-only

Item #2 Understanding 1 pt

Arduino consists of a programmable circuit board and the

A. Arduino IDE ✓ (Correct)
B. Serial Monitor
C. breadboard only

Item #3 Understanding 1 pt

On the Arduino UNO, programs are uploaded from the PC through the

A. USB port ✓ (Correct)
B. power jack
C. reset button

Item #4 Understanding 1 pt

The reset button on the UNO

A. restarts the program from setup() ✓ (Correct)
B. erases the sketch
C. clears the serial monitor

Item #5 Understanding 1 pt

The "brain" of the Arduino UNO that stores and runs the sketch is the

A. microcontroller (ATmega328P) ✓ (Correct)
B. USB chip only
C. voltage regulator

Item #6 Remembering 1 pt

The UNO has how many digital pins (0–13)?

A. 14 ✓ (Correct)
B. 6
C. 20

Item #7 Remembering 1 pt

The UNO has how many analog input pins?

A. 6 (A0–A5) ✓ (Correct)
B. 14
C. 8

Item #8 Understanding 1 pt

Analog pins are used to

A. read varying voltage levels from sensors ✓ (Correct)
B. send only ON/OFF signals
C. power motors directly

Item #9 Remembering 1 pt

analogRead() on the UNO returns values in the range

A. 0 to 1023 ✓ (Correct)
B. 0 to 255
C. 0 to 5

Item #10 Remembering 1 pt

The power pins on the UNO header include

A. 5V, 3.3V, GND, and Vin ✓ (Correct)
B. only 12V
C. USB and HDMI

Item #11 Remembering 1 pt

The UNO has a built-in LED wired to digital pin

A. 13 ✓ (Correct)
B. 7
C. 0

Item #12 Applying 1 pt

Some digital pins marked with a ~ can also produce

A. PWM output ✓ (Correct)
B. analog input
C. Wi-Fi

Item #13 Remembering 1 pt

Which board is the standard beginner board used in this course?

A. Arduino UNO ✓ (Correct)
B. Arduino Mega
C. Pandaboard

Item #14 Remembering 1 pt

Which Arduino board has the most digital pins (54) for larger projects?

A. Arduino Mega ✓ (Correct)
B. Arduino Nano
C. Arduino Leonardo

Item #15 Remembering 1 pt

Which Arduino board is tiny and breadboard-friendly?

A. Arduino Nano ✓ (Correct)
B. Arduino Mega
C. Arduino Due

Item #16 Understanding 1 pt

In the bare minimum sketch, setup() runs

A. once at power-on or reset ✓ (Correct)
B. continuously
C. only when unplugged

Item #17 Understanding 1 pt

In the bare minimum sketch, loop() runs

A. over and over forever ✓ (Correct)
B. only once
C. never

Item #18 Understanding 1 pt

setup() is used to initialize

A. variables and pin modes ✓ (Correct)
B. serial printing only
C. libraries only

Item #19 Understanding 1 pt

A breadboard is used to

A. build temporary circuits without soldering ✓ (Correct)
B. store uploaded programs
C. measure resistance

Item #20 Remembering 1 pt

The longer leg of an LED is the

A. anode (+) ✓ (Correct)
B. cathode (–)
C. gate

Item #21 Understanding 1 pt

A pushbutton on a digital pin acts as a

A. digital input ✓ (Correct)
B. actuator
C. display

Item #22 Understanding 1 pt

A potentiometer is read by the Arduino as an

A. analog value ✓ (Correct)
B. digital pulse
C. serial packet

Item #23 Applying 1 pt

The TMP36 temperature sensor outputs a voltage

A. proportional to temperature ✓ (Correct)
B. that is always 5V
C. proportional to light

Item #24 Applying 1 pt

A servo motor is an example of a

A. sensor
B. actuator ✓ (Correct)
C. resistor

Worksheet 2: Quiz — Lab Activity 1: Blinking LED (24 Items)

Item #1 Remembering 1 pt

Laboratory Activity 1 builds a blinking LED using which board?

A. Arduino UNO ✓ (Correct)
B. Raspberry Pi
C. Mbed

Item #2 Remembering 1 pt

The lab connects the LED in series with a

A. 1 kΩ resistor ✓ (Correct)
B. 10 kΩ potentiometer
C. capacitor

Item #3 Remembering 1 pt

The color code for the lab's 1 kΩ resistor is

A. brown-black-red ✓ (Correct)
B. red-red-brown
C. yellow-violet-red

Item #4 Remembering 1 pt

How many male-to-male jumper wires does the lab require?

A. 2 ✓ (Correct)
B. 4
C. 1

Item #5 Remembering 1 pt

How many LEDs does the circuit use?

A. 1 ✓ (Correct)
B. 2
C. 3

Item #6 Understanding 1 pt

One jumper wire carries the signal from a digital pin; the other connects to

A. GND ✓ (Correct)
B. 5V
C. 3.3V

Item #7 Understanding 1 pt

The LED and the resistor are connected

A. in series ✓ (Correct)
B. in parallel
C. back-to-back

Item #8 Understanding 1 pt

The purpose of the current-limiting resistor is to

A. protect the LED from excessive current ✓ (Correct)
B. make the LED blink faster
C. store charge

Item #9 Understanding 1 pt

Connecting an LED directly to 5 V without a resistor will

A. likely burn out the LED ✓ (Correct)
B. do nothing
C. dim the LED forever

Item #10 Understanding 1 pt

The longer LED leg (anode) connects toward the

A. resistor/pin side ✓ (Correct)
B. ground side
C. either side

Item #11 Remembering 1 pt

The Blink example sketch lives under File → Examples →

A. 01.Basics ✓ (Correct)
B. 03.Analog
C. 05.Characters

Item #12 Remembering 1 pt

Before uploading, the IDE board setting must be

A. Arduino Uno ✓ (Correct)
B. Arduino Mega
C. Generic AVR

Item #13 Remembering 1 pt

The correct board port is selected under Tools →

A. Port ✓ (Correct)
B. Monitor
C. Library

Item #14 Remembering 1 pt

The Upload button in the IDE is the

A. right-arrow (→) button ✓ (Correct)
B. left-arrow button
C. checkmark only

Item #15 Understanding 1 pt

In the stock Blink sketch, the LED stays ON for

A. 1000 ms (1 second) ✓ (Correct)
B. 250 ms
C. 5000 ms

Item #16 Understanding 1 pt

A full blink cycle (ON + OFF) in the stock sketch takes

A. 2 seconds ✓ (Correct)
B. 1 second
C. 4 seconds

Item #17 Applying 1 pt

Changing both delays to delay(250) makes the LED blink

A. 4 times faster ✓ (Correct)
B. slower
C. the same

Item #18 Applying 1 pt

digitalWrite(pin, HIGH) on the LED pin makes the LED

A. turn ON ✓ (Correct)
B. turn OFF
C. blink twice

Item #19 Applying 1 pt

If the LED never lights, which is a possible cause?

A. LED legs reversed in the breadboard ✓ (Correct)
B. delay is 1000 ms
C. the board was uploaded successfully

Item #20 Understanding 1 pt

The circuit ground wire must connect from the breadboard to any

A. GND pin on the Arduino ✓ (Correct)
B. analog pin
C. USB pin

Item #21 Understanding 1 pt

Which function in the Blink sketch turns the LED on and off repeatedly?

A. loop() ✓ (Correct)
B. setup()
C. main()

Item #22 Understanding 1 pt

The delay() function is used to

A. pause the program for a given time ✓ (Correct)
B. read a sensor
C. upload a sketch

Item #23 Applying 1 pt

One learning outcome of the lab is to troubleshoot common

A. wiring or connection errors ✓ (Correct)
B. IDE downloads
C. Windows updates

Item #24 Applying 1 pt

Before touching the breadboard, the lab procedure says to

A. disconnect the Arduino from power/USB ✓ (Correct)
B. hold the LED by its legs
C. turn the potentiometer

Worksheet 3: Quiz — Arduino Part 2 (26 Items)

Item #1 Remembering 1 pt

Arduino is an _____ electronic prototyping platform.

A. open-source ✓ (Correct)
B. closed-source
C. commercial-only

Item #2 Understanding 1 pt

In the bare minimum code, which function runs only once when the Arduino is powered on or reset?

A. setup() ✓ (Correct)
B. loop()
C. main()

Item #3 Understanding 1 pt

The setup() function is used to initialize

A. variables and pin modes ✓ (Correct)
B. serial printing only
C. millis timers

Item #4 Understanding 1 pt

The loop() function runs

A. continuously until power is off ✓ (Correct)
B. only once
C. only when reset

Item #5 Understanding 1 pt

In microcontroller programming, loop() is similar to which construct?

A. while (1) ✓ (Correct)
B. if (1)
C. switch (1)

Item #6 Understanding 1 pt

The main logic of an Arduino sketch goes in the

A. loop() ✓ (Correct)
B. setup()
C. libraries folder

Item #7 Understanding 1 pt

Libraries are collections of code that make it easy to connect to a

A. sensor, display, or module ✓ (Correct)
B. power supply only
C. USB cable

Item #8 Remembering 1 pt

Which built-in library makes it easy to talk to character LCD displays?

A. LiquidCrystal ✓ (Correct)
B. Servo
C. EEPROM

Item #9 Applying 1 pt

Hundreds of additional Arduino libraries are available

A. for download on the Internet ✓ (Correct)
B. only on CD
C. inside the board

Item #10 Understanding 1 pt

A sensor is a converter that measures a physical quantity and converts it to a

A. signal readable by an instrument or Arduino ✓ (Correct)
B. mechanical motion
C. battery charge

Item #11 Remembering 1 pt

Which of these is a sensor?

A. Temperature sensor ✓ (Correct)
B. Engine
C. Valve

Item #12 Understanding 1 pt

An actuator converts energy into

A. motion ✓ (Correct)
B. light
C. sound only

Item #13 Remembering 1 pt

Which of these is an actuator?

A. Pump ✓ (Correct)
B. Potentiometer
C. Thermistor

Item #14 Understanding 1 pt

Sensors and actuators can be either

A. digital or analog ✓ (Correct)
B. USB or serial
C. 5V or 3V only

Item #15 Understanding 1 pt

Many sensors and actuators for Arduino come with

A. ready-made libraries ✓ (Correct)
B. no documentation
C. custom PCBs only

Item #16 Understanding 1 pt

A potentiometer is a simple knob that provides a variable

A. resistance ✓ (Correct)
B. capacitance
C. voltage source

Item #17 Understanding 1 pt

The Arduino reads a potentiometer as an

A. analog value ✓ (Correct)
B. digital pulse
C. serial packet

Item #18 Remembering 1 pt

Which of these is an example of a single-board device from the lecture?

A. Raspberry Pi ✓ (Correct)
B. Hard disk
C. Monitor

Item #19 Remembering 1 pt

Mbed is an example of a

A. single-board device ✓ (Correct)
B. resistor code
C. sensor library

Item #20 Understanding 1 pt

Early microcomputers typically consisted of circuit boards plugged into a

A. backplane ✓ (Correct)
B. breadboard
C. USB hub

Item #21 Remembering 1 pt

For the traffic light individual assessment, how many 220 Ω resistors are needed?

A. 3 ✓ (Correct)
B. 1
C. 5

Item #22 Remembering 1 pt

The traffic light assessment builds with which board?

A. Arduino UNO ✓ (Correct)
B. Raspberry Pi
C. Pandaboard

Item #23 Remembering 1 pt

Which LEDs does the traffic light assessment require?

A. Red, Yellow, and Green ✓ (Correct)
B. Red and Blue only
C. White and Green

Item #24 Remembering 1 pt

The traffic light circuit is created in

A. Tinkercad ✓ (Correct)
B. LibreOffice
C. the Serial Monitor

Item #25 Creating / Synthesis 5 pts (Simulation) Required

Simulation Task A — Traffic Light (5 pts). In the Virtual Arduino Lab, build the traffic light: red, yellow, and green LEDs, each through a 220 Ω resistor, to digital pins 8, 9, and 10 with a common ground. Write a sketch that cycles red for 2 seconds, yellow for 1 second, and green for 2 seconds, repeating.

Analytic Scoring Rubric:
Rubric: Three LEDs wired to pins 8 (red), 9 (yellow), and 10 (green), each with a 220 Ω resistor, common ground | Keywords: pinMode, digitalWrite, delay, 2000, 1000, red-yellow-green sequence, loop

Item #26 Creating / Synthesis 5 pts (Simulation) Required

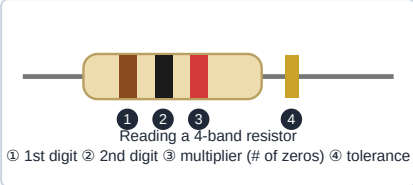
Simulation Task B — Button-Controlled LED with Serial Monitor (5 pts). In the Virtual Arduino Lab, wire a pushbutton to digital pin 2 (using INPUT_PULLUP) and an LED through a 220 Ω resistor to pin 13, common ground. Write a sketch that lights the LED while the button is pressed and prints the button state to the serial monitor.

Analytic Scoring Rubric:
Rubric: Pushbutton on pin 2 configured INPUT_PULLUP, LED on pin 13; sketch reads the button each loop(), drives the LED from its state, and prints the state over Serial | Keywords: pinMode INPUT_PULLUP, digitalRead, digitalWrite, Serial.print, if statement, loop

Worksheet 4: Quiz — Resistor Color Code (24 Items)

Item #1 Remembering 1 pt

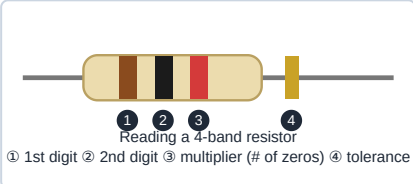
On a 4-band resistor, the 1st band color gives the



A. 1st number ✓ (Correct)
B. tolerance
C. multiplier only

Item #2 Remembering 1 pt

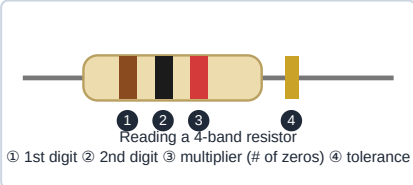
On a 4-band resistor, the 2nd band color gives the



A. 2nd number ✓ (Correct)
B. # of zeros
C. tolerance

Item #3 Remembering 1 pt

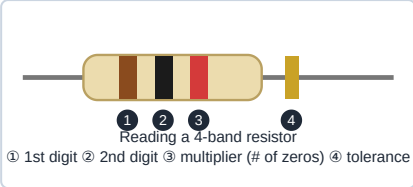
On a 4-band resistor, the 3rd band color gives the



A. # of zeros (multiplier) ✓ (Correct)
B. 1st number
C. tolerance

Item #4 Remembering 1 pt

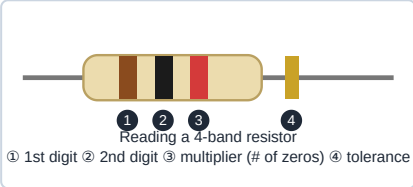
On a 4-band resistor, the 4th band color gives the



A. tolerance (±) ✓ (Correct)
B. 3rd digit
C. voltage rating

Item #5 Remembering 1 pt

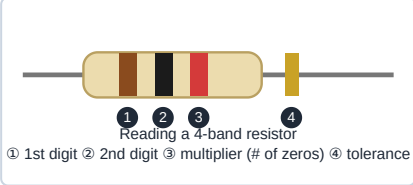
To read a resistor, hold it with the gold or silver tolerance band to the



A. right ✓ (Correct)
B. left
C. bottom

Item #6 Remembering 1 pt

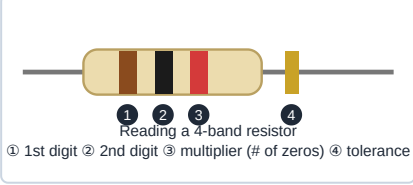
A gold tolerance band means ±



A. 5% ✓ (Correct)
B. 10%
C. 2%

Item #7 Remembering 1 pt

A silver tolerance band means ±



A. 10% ✓ (Correct)
B. 5%
C. 1%

Item #8 Applying 1 pt

Red, Black, Yellow with silver tolerance (lecture example) equals



A. 200,000 Ω ± 10% ✓ (Correct)
B. 20,000 Ω ± 10%
C. 2,000 Ω ± 10%

Item #9 Applying 1 pt

Brown, Black, Red, Gold (Example 1) has a nominal value of



A. 1,000 Ω ✓ (Correct)
B. 100 Ω
C. 10,000 Ω

Item #10 Applying 1 pt

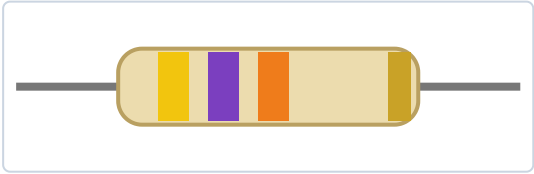
The tolerance in Example 1 (gold band) is



A. ±5% ✓ (Correct)
B. ±10%
C. ±2%

Item #11 Applying 1 pt

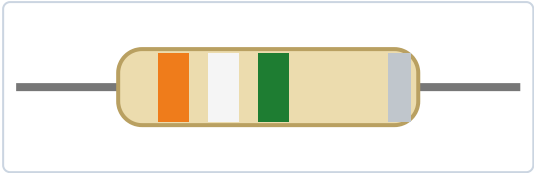
Yellow, Violet, Orange, Gold (Example 2) equals



A. 47 kΩ ± 5% ✓ (Correct)
B. 470 kΩ ± 5%
C. 4.7 kΩ ± 5%

Item #12 Applying 1 pt

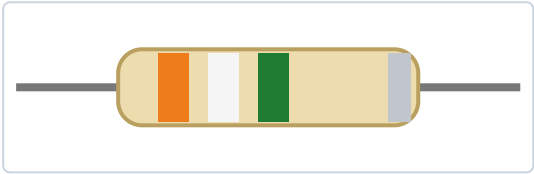
Orange, White, Green, Silver (Example 3) equals



A. 3.9 MΩ ± 10% ✓ (Correct)
B. 390 kΩ ± 10%
C. 3.9 kΩ ± 10%

Item #13 Applying 1 pt

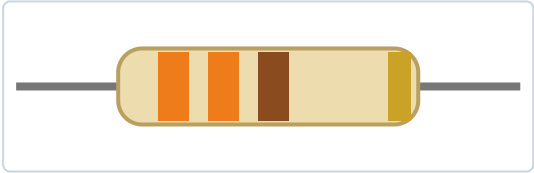
The tolerance in Example 3 (silver band) is



A. ±10% ✓ (Correct)
B. ±5%
C. ±1%

Item #14 Applying 1 pt

Orange, Orange, Brown, Gold (lecture practice) equals



A. 330 Ω ± 5% ✓ (Correct)
B. 33 Ω ± 5%
C. 3.3 kΩ ± 5%

Item #15 Understanding 1 pt

5-band color codes are used for resistors with tolerances of

A. ±1% or ±2% ✓ (Correct)
B. ±5% or ±10%
C. ±20% or ±25%

Item #16 Understanding 1 pt

In a 5-band code, the first three bands are

A. significant digits ✓ (Correct)
B. multiplier and tolerance
C. all multipliers

Item #17 Understanding 1 pt

In a 5-band code, the 4th band is the

A. multiplier ✓ (Correct)
B. tolerance
C. 1st digit

Item #18 Applying 1 pt

5-band Brown, Black, Black, Red, Brown (Example 4) equals



A. 1 kΩ ± 2% ✓ (Correct)
B. 100 Ω ± 2%
C. 10 kΩ ± 2%

Item #19 Applying 1 pt

5-band Blue, Grey, Black, Orange, Brown (Example 5) equals



A. 680 kΩ ± 1% ✓ (Correct)
B. 68 kΩ ± 1%
C. 6.8 MΩ ± 1%

Item #20 Remembering 1 pt

Resistors are color coded for

A. easy reading ✓ (Correct)
B. higher voltage
C. heat resistance

Item #21 Remembering 1 pt

To decode bands you match each color to its

A. color code chart ✓ (Correct)
B. serial monitor
C. multimeter dial

Item #22 Remembering 1 pt

The color yellow represents which digit?

A. 4 ✓ (Correct)
B. 5
C. 6

Item #23 Remembering 1 pt

The color violet represents which digit?

A. 7 ✓ (Correct)
B. 8
C. 2

Item #24 Understanding 1 pt

A multimeter is used to check whether the color-coded value

A. matches the measured resistance ✓ (Correct)
B. equals the voltage
C. matches the current

Worksheet 5: Quiz — Programming with Arduino (24 Items)

Item #1 Remembering 1 pt

The guide is divided into 5 parts. Part 2 is

A. Basic Electronics ✓ (Correct)
B. Introduction to Arduino
C. Arduino Examples

Item #2 Remembering 1 pt

Which part of the guide covers the Arduino Development Environment?

A. Part 3 ✓ (Correct)
B. Part 1
C. Part 5

Item #3 Remembering 1 pt

Part 5 of the guide covers

A. Arduino Examples ✓ (Correct)
B. Basic Electronics
C. the C++ language

Item #4 Understanding 1 pt

In which part do you learn about LEDs and resistors?

A. Basic Electronics ✓ (Correct)
B. Arduino Examples
C. Introduction

Item #5 Understanding 1 pt

Connecting both terminals of a battery directly together is called a

A. short circuit ✓ (Correct)
B. open loop
C. solder joint

Item #6 Understanding 1 pt

A breadboard is used to

A. wire circuits without soldering ✓ (Correct)
B. store programs
C. measure resistance

Item #7 Remembering 1 pt

The Arduino is connected to the PC for programming using a

A. USB cable ✓ (Correct)
B. HDMI cable
C. ethernet cable

Item #8 Remembering 1 pt

The board's restart (reset) button

A. restarts the program from setup() ✓ (Correct)
B. clears the code
C. erases all libraries

Item #9 Understanding 1 pt

In the IDE, the check/verify step

A. compiles the code and reports errors ✓ (Correct)
B. uploads to the board
C. opens the Serial Monitor

Item #10 Understanding 1 pt

Uploading sends the compiled program to the board's

A. microcontroller ✓ (Correct)
B. hard drive
C. USB hub

Item #11 Understanding 1 pt

The Serial Monitor is used to

A. view data sent from the board ✓ (Correct)
B. draw circuits
C. install libraries

Item #12 Understanding 1 pt

For Serial communication to work, the monitor's baud rate must match the

A. baud rate in Serial.begin() ✓ (Correct)
B. board voltage
C. pin number

Item #13 Remembering 1 pt

To turn an LED on with a digital pin you use

A. digitalWrite(pin, HIGH) ✓ (Correct)
B. digitalRead(pin)
C. analogWrite(pin, 1023)

Item #14 Remembering 1 pt

To read the state of a switch you use

A. digitalRead() ✓ (Correct)
B. analogWrite()
C. tone()

Item #15 Applying 1 pt

A potentiometer example reads its position with

A. analogRead() ✓ (Correct)
B. digitalRead()
C. pulseIn()

Item #16 Understanding 1 pt

analogRead() on an Arduino Uno returns values from

A. 0 to 1023 ✓ (Correct)
B. 0 to 255
C. 0 to 5

Item #17 Understanding 1 pt

analogWrite() on an Arduino Uno accepts values from

A. 0 to 255 ✓ (Correct)
B. 0 to 1023
C. 0 to 100

Item #18 Understanding 1 pt

analogWrite() is used to produce

A. PWM output ✓ (Correct)
B. serial data
C. analog input

Item #19 Applying 1 pt

The temperature sensor example outputs a voltage

A. proportional to temperature ✓ (Correct)
B. that is always 5V
C. proportional to current only

Item #20 Applying 1 pt

A light sensor (photoresistor/LDR) changes its resistance with

A. light level ✓ (Correct)
B. temperature
C. humidity

Item #21 Applying 1 pt

A thermistor is a resistor whose resistance depends on

A. temperature ✓ (Correct)
B. light
C. magnetic fields

Item #22 Understanding 1 pt

Which example from the guide blinks an LED using delay()?

A. Blinking LED ✓ (Correct)
B. Thermistor
C. LCD display

Item #23 Remembering 1 pt

A variable of type float stores

A. decimal numbers ✓ (Correct)
B. only whole numbers
C. text strings

Item #24 Remembering 1 pt

A variable of type int stores

A. whole numbers ✓ (Correct)
B. decimal numbers
C. characters only

Summative Assessment: 1st Quarter Examination (77 Items)

Item #1 Applying / Analysis 1 pt

Which one is the fastest Led blinking?

A. delay(500); ✓ (Correct)
B. delay(1000);
C. delay(1500);

Item #2 Applying / Analysis 1 pt

Which one of the following is a user defined function?

A. loop();
B. calculate(); ✓ (Correct)
C. setup();

Item #3 Applying / Analysis 1 pt

Which function runs only one time?

A. loop();
B. calculate();
C. setup(); ✓ (Correct)

Item #4 Applying / Analysis 1 pt

To pause a system for 2 seconds in Arduino, we use which line of code?

A. delay(20);
B. delay(200);
C. delay(2000); ✓ (Correct)

Item #5 Applying / Analysis 1 pt

How many analog pins are on an Arduino board?

A. 5
B. 6 ✓ (Correct)
C. 7

Item #6 Applying / Analysis 1 pt

How many digital pins are on an Arduino board?

A. 10
B. 12
C. 14 ✓ (Correct)

Item #7 Applying / Analysis 1 pt

All of these are input devices EXCEPT

A. Temp Sensor
B. LED ✓ (Correct)
C. Ultrasonic Sensor

Item #8 Applying / Analysis 1 pt

What is the result of the following code: print(7%3);

A. 0
B. 1 ✓ (Correct)
C. 2

Item #9 Applying / Analysis 1 pt

The following line of code means pinMode(12, OUTPUT);

A. Set the device in pin 12 as output. ✓ (Correct)
B. Set the device in pin 12 as input.
C. Set the device in pin 12 set to Off state.

Item #10 Applying / Analysis 1 pt

To control the current through a circuit we use

A. Wire
B. Sensor
C. Resistor ✓ (Correct)

Item #11 Applying / Analysis 1 pt

Code writing and uploading into the Arduino board is done using which tool?

A. IDE ✓ (Correct)
B. PC
C. SOME

Item #12 Applying / Analysis 1 pt

The Arduino board is also called a

A. Timer
B. Oscillator
C. Microcontroller ✓ (Correct)

Item #13 Applying / Analysis 1 pt

To ground the circuit, which pin is used in Arduino?

A. RESET
B. GND ✓ (Correct)
C. PIN

Item #14 Applying / Analysis 1 pt

Arduino IDE consists of 2 functions. What are they?

A. build() & loop()
B. setup() & build()
C. setup() & loop() ✓ (Correct)

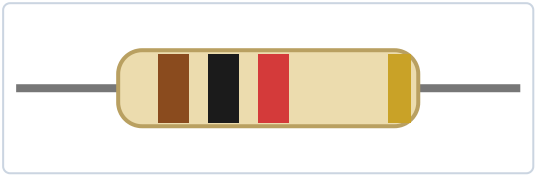
Item #15 Applying / Analysis 1 pt

digitalWrite(13, LOW); turns the LED on pin 13

A. ON
B. OFF ✓ (Correct)
C. DIM

Item #16 Applying (Decoding) 1 pt

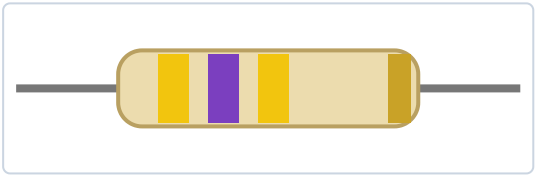
Decode the value: Brown, Black, Red, Gold.



A. 1 kΩ ±5% ✓ (Correct)
B. 100 Ω ±5%
C. 10 kΩ ±5%

Item #17 Applying (Decoding) 1 pt

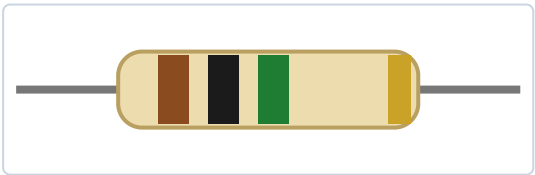
Decode the value: Yellow, Violet, Yellow, Gold.



A. 470 kΩ ±5% ✓ (Correct)
B. 47 kΩ ±5%
C. 4.7 MΩ ±5%

Item #18 Applying (Decoding) 1 pt

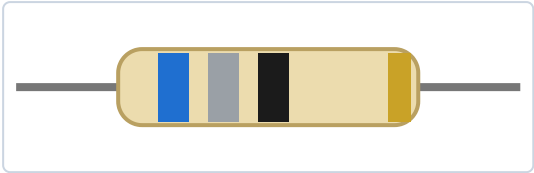
Decode the value: Brown, Black, Green, Gold.



A. 1 MΩ ±5% ✓ (Correct)
B. 100 kΩ ±5%
C. 10 MΩ ±5%

Item #19 Applying (Decoding) 1 pt

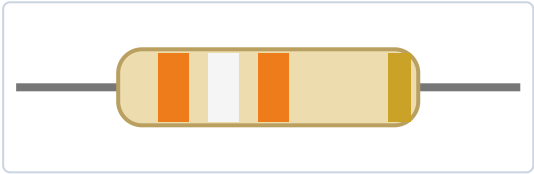
Decode the value: Blue, Grey, Black, Gold.



A. 68 Ω ±5% ✓ (Correct)
B. 680 Ω ±5%
C. 6.8 kΩ ±5%

Item #20 Applying (Decoding) 1 pt

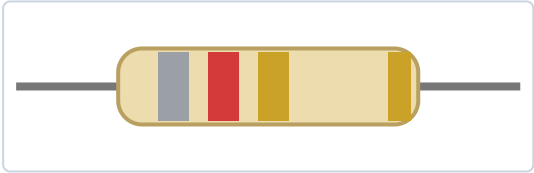
Decode the value: Orange, White, Orange, Gold.



A. 39 kΩ ±5% ✓ (Correct)
B. 390 kΩ ±5%
C. 3.9 kΩ ±5%

Item #21 Applying (Decoding) 1 pt

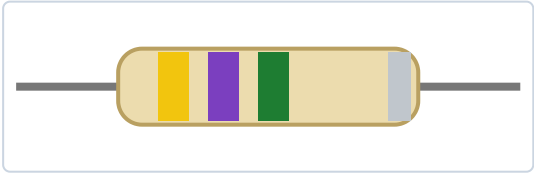
Decode the value: Grey, Red, Gold, Gold.



A. 8.2 Ω ±5% ✓ (Correct)
B. 82 Ω ±5%
C. 820 Ω ±5%

Item #22 Applying (Decoding) 1 pt

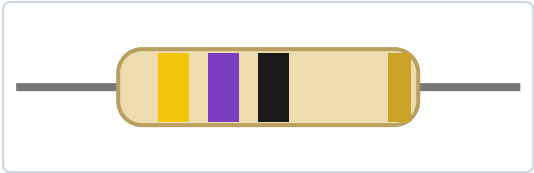
Decode the value: Yellow, Violet, Green, Silver.



A. 4.7 MΩ ±10% ✓ (Correct)
B. 47 MΩ ±10%
C. 4.7 kΩ ±10%

Item #23 Applying (Decoding) 1 pt

Decode the value: Yellow, Violet, Black, Gold.



A. 47 Ω ±5% ✓ (Correct)
B. 470 Ω ±5%
C. 4.7 kΩ ±5%

Item #24 Applying (Decoding) 1 pt

Decode the value: Orange, Black, Red, Silver.



A. 3 kΩ ±10% ✓ (Correct)
B. 30 kΩ ±10%
C. 300 Ω ±10%

Item #25 Applying (Decoding) 1 pt

Decode the value: Green, Blue, Yellow, Silver.



A. 560 kΩ ±10% ✓ (Correct)
B. 56 kΩ ±10%
C. 5.6 MΩ ±10%

Item #26 Applying (Computation) 1 pt

A 12 V source pushes 60 mA through a resistor. What is its resistance?

A. 200 Ω ✓ (Correct)
B. 2 kΩ
C. 20 Ω

Item #27 Applying (Computation) 1 pt

What resistance draws 10 mA from a 5 V source?

A. 500 Ω ✓ (Correct)
B. 5 kΩ
C. 50 Ω

Item #28 Applying (Computation) 1 pt

What resistance draws 33 mA from a 3.3 V source?

A. 10 Ω
B. 100 Ω ✓ (Correct)
C. 1 kΩ

Item #29 Applying (Computation) 1 pt

What resistance draws 80 mA from a 24 V source?

A. 30 Ω
B. 300 Ω ✓ (Correct)
C. 3 kΩ

Item #30 Applying (Computation) 1 pt

What resistance draws 30 mA from a 1.5 V source?

A. 5 Ω
B. 50 Ω ✓ (Correct)
C. 500 Ω

Item #31 Applying (Computation) 1 pt

What resistance draws 40 mA from a 6 V source?

A. 15 Ω
B. 150 Ω ✓ (Correct)
C. 1.5 kΩ

Item #32 Applying (Computation) 1 pt

What resistance draws 55 mA from a 110 V source?

A. 200 Ω
B. 2 kΩ ✓ (Correct)
C. 20 kΩ

Item #33 Applying (Computation) 1 pt

What resistance draws 2 mA from a 3 V source?

A. 150 Ω
B. 1.5 kΩ ✓ (Correct)
C. 15 kΩ

Item #34 Applying (Computation) 1 pt

What resistance draws 50 mA from a 20 V source?

A. 40 Ω
B. 400 Ω ✓ (Correct)
C. 4 kΩ

Item #35 Applying (Computation) 1 pt

What resistance draws 64 mA from a 16 V source?

A. 25 Ω
B. 250 Ω ✓ (Correct)
C. 2.5 kΩ

Item #36 Remembering (Recall) 1 pt

It is an open-source electronics platform based on easy-to-use hardware and software.

Answer Key: `arduino`

Item #37 Remembering (Recall) 1 pt

The tiny computer on the Arduino board is called a _____.

Answer Key: `microcontroller`

Item #38 Remembering (Recall) 1 pt

A solderless board used to build temporary prototypes without soldering.

Answer Key: `breadboard`

Item #39 Remembering (Recall) 1 pt

Wires with a pin on both ends used to connect components on a breadboard.

Accepted Identification Key Variants: `jumper wires | jumper wire | jumper`

Item #40 Remembering (Recall) 1 pt

A three-terminal resistor with a rotating shaft used to adjust resistance.

Accepted Identification Key Variants: `potentiometer | trimmer | trimpot`

Item #41 Remembering (Recall) 1 pt

This component allows electricity to flow in only one direction.

Accepted Identification Key Variants: `diode | signal diode | rectifier diode`

Item #42 Remembering (Recall) 1 pt

The LED's longer leg is called the _____.

Answer Key: `anode`

Item #43 Remembering (Recall) 1 pt

The LED's shorter leg is called the _____.

Answer Key: `cathode`

Item #44 Remembering (Recall) 1 pt

How many ohms is a 10 kΩ resistor? Write digits only.

Accepted Identification Key Variants: `10000 | 10,000`

Item #45 Remembering (Recall) 1 pt

The instrument used to measure electrical resistance is called a _____.

Accepted Identification Key Variants: `multimeter | ohmmeter | multi meter`

Item #46 Remembering (Recall) 1 pt

Color code: a resistor with bands Red-Red-Brown-Gold is how many ohms? (digits only)

Answer Key: `220`

Item #47 Remembering (Recall) 1 pt

The SIK sensor that measures nearby object distance using sound waves is the _____ sensor.

Accepted Identification Key Variants: `ultrasonic | sonar`

Item #48 Remembering (Recall) 1 pt

The SIK temperature sensor (TMP36) outputs a voltage proportional to _____.

Accepted Identification Key Variants: `temperature | heat`

Item #49 Remembering (Recall) 1 pt

A light-dependent resistor changes its resistance based on _____.

Accepted Identification Key Variants: `light | brightness | light intensity`

Item #50 Remembering (Recall) 1 pt

This SIK component converts electrical energy into continuous mechanical motion.

Accepted Identification Key Variants: `motor | dc motor | direct current motor`

Item #51 Remembering (Recall) 1 pt

A programmable light that can glow in many colors using three channels is called a _____ LED.

Answer Key: `rgb`

Item #52 Remembering (Recall) 1 pt

The pin used to reset the Arduino board is the _____ pin.

Answer Key: `reset`

Item #53 Remembering (Recall) 1 pt

The pin that provides 0 volts reference is the _____ pin.

Accepted Identification Key Variants: `gnd | ground`

Item #54 Remembering (Recall) 1 pt

The pin that provides regulated +5V on an Arduino Uno is the _____ pin.

Accepted Identification Key Variants: `5v | 5v power | power`

Item #55 Remembering (Recall) 1 pt

A semiconductor component that amplifies or switches electronic signals is a _____.

Answer Key: `transistor`

Item #56 Understanding 1 pt

Arduino Uno digital pin 13 has a built-in _____.

A. LED ✓ (Correct)
B. button
C. motor

Item #57 Understanding 1 pt

Which of these is a function used to send text over the serial port?

A. Serial.print() ✓ (Correct)
B. delay()
C. pinMode()

Item #58 Understanding 1 pt

analogRead() returns values in which range?

A. 0 to 1023 ✓ (Correct)
B. 0 to 255
C. 0 to 5

Item #59 Understanding 1 pt

digitalWrite() can set a digital pin to

A. HIGH or LOW ✓ (Correct)
B. 0 to 1023
C. positive or negative

Item #60 Understanding 1 pt

A baud rate commonly used with Serial.begin() is

A. 9600 ✓ (Correct)
B. 1200
C. 96000

Item #61 Understanding 1 pt

The loop() function in Arduino

A. runs repeatedly ✓ (Correct)
B. runs once
C. never runs

Item #62 Understanding 1 pt

With pinMode(pin, INPUT_PULLUP), a pushbutton reads _____ when pressed.

A. LOW ✓ (Correct)
B. HIGH
C. 5 volts

Item #63 Understanding 1 pt

Which unit measures electrical resistance?

A. ohm ✓ (Correct)
B. volt
C. ampere

Item #64 Understanding 1 pt

The Arduino Uno receives power over USB at approximately how many volts?

A. 5 ✓ (Correct)
B. 12
C. 9

Item #65 Understanding 1 pt

Which Arduino function configures a pin as an output?

A. pinMode() ✓ (Correct)
B. analogRead()
C. map()

Item #66 Remembering (Visual ID) 1 pt Required

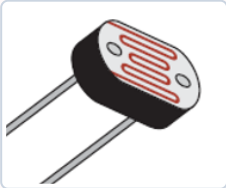
Identify the component shown in the picture.



Accepted Identification Key Variants: `arduino uno | arduino | arduino board | uno board | uno`

Item #67 Remembering (Visual ID) 1 pt Required

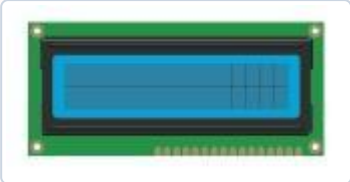
Identify the component shown in the picture.



Accepted Identification Key Variants: `ldr | light dependent resistor | photoresistor | light-dependent resistor`

Item #68 Remembering (Visual ID) 1 pt Required

Identify the component shown in the picture.



Accepted Identification Key Variants: `lcd | lcd display | liquid crystal display | 16x2 lcd | character lcd`

Item #69 Remembering (Visual ID) 1 pt Required

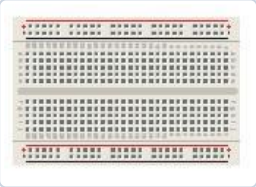
Identify the component shown in the picture.



Accepted Identification Key Variants: `buzzer | piezo buzzer | piezo | sound module`

Item #70 Remembering (Visual ID) 1 pt Required

Identify the component shown in the picture.



Accepted Identification Key Variants: `breadboard | solderless breadboard`

Item #71 Remembering (Visual ID) 1 pt Required

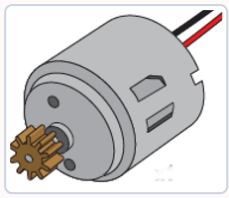
Identify the component shown in the picture.



Accepted Identification Key Variants: `led | light emitting diode | light-emitting diode | leds`

Item #72 Remembering (Visual ID) 1 pt Required

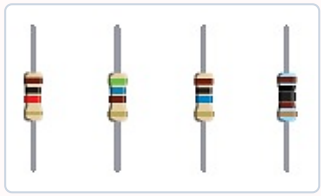
Identify the component shown in the picture.



Accepted Identification Key Variants: `dc motor | motor | direct current motor`

Item #73 Remembering (Visual ID) 1 pt Required

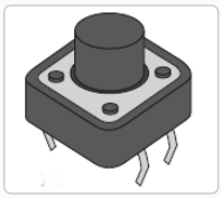
Identify the component shown in the picture.



Accepted Identification Key Variants: `resistor | resistors`

Item #74 Remembering (Visual ID) 1 pt Required

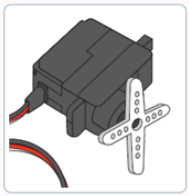
Identify the component shown in the picture.



Accepted Identification Key Variants: `pushbutton | push button | push button switch | button | tactile switch`

Item #75 Remembering (Visual ID) 1 pt Required

Identify the component shown in the picture.



Accepted Identification Key Variants: `servo | servo motor | sg90 | servo motor sg90`

Item #76 Creating / Synthesis 5 pts (Lab Sketch) Required

Part IV — Problem Solving (5 pts). Write a complete Arduino sketch that will turn an LED on and off using a push button. Build and test your circuit in the Virtual Arduino Lab below, then paste your final code here.

Analytic Scoring Rubric:
Rubric: A working sketch that sets pinMode() for the pushbutton (INPUT_PULLUP or an external pull-down) and the LED pin as OUTPUT, reads the button inside loop(), and uses digitalWrite() to turn the LED ON and OFF. | Keywords: pinmode; digitalwrite; loop; button=pushbutton, switch; led

Item #77 Creating / Synthesis | 10 pts (Lab Circuit) | Required

Part IV — Problem Solving (10 pts). Build the Arduino circuit that turns an LED on and off using a push button. Construct it yourself on the simulation canvas in the Virtual Arduino Lab below, then describe your connections here — which Arduino pins you used and where each component goes.

Analytic Scoring Rubric:
Rubric: A correct circuit: Arduino wired to a breadboard, LED with a series current-limiting resistor (220 Ω to 1 kΩ), pushbutton between a digital pin and GND (INPUT_PULLUP or pull-down), LED anode to a digital pin and cathode to GND, all grounds common. | Keywords: breadboard; resistor; button=pushbutton, switch; gnd=ground, gnd; pin=pin, pinmode, digital; led

SECTION 4: CONSOLIDATED MASTER ANSWER KEY

Quick-reference verification matrix for instructors, proctors, and examiners.

Worksheet 1: Quiz — Arduino Part 1 (24 Items)

#	Correct Answer / Key	#	Correct Answer / Key
1	open-source	13	Arduino UNO
2	Arduino IDE	14	Arduino Mega
3	USB port	15	Arduino Nano
4	restarts the program from setup()	16	once at power-on or reset
5	microcontroller (ATmega328P)	17	over and over forever
6	14	18	variables and pin modes
7	6 (A0–A5)	19	build temporary circuits without soldering
8	read varying voltage levels from sensors	20	anode (+)
9	0 to 1023	21	digital input
10	5V, 3.3V, GND, and Vin	22	analog value
11	13	23	proportional to temperature
12	PWM output	24	actuator

Worksheet 2: Quiz — Lab Activity 1: Blinking LED (24 Items)

#	Correct Answer / Key	#	Correct Answer / Key
1	Arduino UNO	13	Port
2	1 kΩ resistor	14	right-arrow (→) button
3	brown-black-red	15	1000 ms (1 second)
4	2	16	2 seconds
5	1	17	4 times faster
6	GND	18	turn ON
7	in series	19	LED legs reversed in the breadboard
8	protect the LED from excessive current	20	GND pin on the Arduino
9	likely burn out the LED	21	loop()
10	resistor/pin side	22	pause the program for a given time
11	01.Basics	23	wiring or connection errors
12	Arduino Uno	24	disconnect the Arduino from power/USB

Worksheet 3: Quiz — Arduino Part 2 (26 Items)

#	Correct Answer / Key	#	Correct Answer / Key
1	open-source	14	digital or analog
2	setup()	15	ready-made libraries
3	variables and pin modes	16	resistance
4	continuously until power is off	17	analog value
5	while (1)	18	Raspberry Pi
6	loop()	19	single-board device
7	sensor, display, or module	20	backplane
8	LiquidCrystal	21	3
9	for download on the Internet	22	Arduino UNO
10	signal readable by an instrument or Arduino	23	Red, Yellow, and Green
11	Temperature sensor	24	Tinkercad
12	motion	25	Rubric: Three LEDs wired to pins 8 (red), 9 (yello...
13	Pump	26	Rubric: Pushbutton on pin 2 configured INPUT_PULLU...

Worksheet 4: Quiz — Resistor Color Code (24 Items)

#	Correct Answer / Key	#	Correct Answer / Key
1	1st number	13	±10%
2	2nd number	14	330 Ω ± 5%
3	# of zeros (multiplier)	15	±1% or ±2%
4	tolerance (±)	16	significant digits
5	right	17	multiplier
6	5%	18	1 kΩ ± 2%
7	10%	19	680 kΩ ± 1%
8	200,000 Ω ± 10%	20	easy reading
9	1,000 Ω	21	color code chart
10	±5%	22	4
11	47 kΩ ± 5%	23	7
12	3.9 MΩ ± 10%	24	matches the measured resistance

Worksheet 5: Quiz — Programming with Arduino (24 Items)

#	Correct Answer / Key	#	Correct Answer / Key
1	Basic Electronics	13	digitalWrite(pin, HIGH)
2	Part 3	14	digitalRead()
3	Arduino Examples	15	analogRead()
4	Basic Electronics	16	0 to 1023
5	short circuit	17	0 to 255
6	wire circuits without soldering	18	PWM output
7	USB cable	19	proportional to temperature
8	restarts the program from setup()	20	light level
9	compiles the code and reports errors	21	temperature
10	microcontroller	22	Blinking LED
11	view data sent from the board	23	decimal numbers
12	baud rate in Serial.begin()	24	whole numbers

Summative Assessment: 1st Quarter Examination (77 Items)

#	Correct Answer / Key	#	Correct Answer / Key
1	delay(500);	40	potentiometer (or variants)
2	calculate();	41	diode (or variants)
3	setup();	42	anode
4	delay(2000);	43	cathode
5	6	44	10000 (or variants)
6	14	45	multimeter (or variants)
7	LED	46	220
8	1	47	ultrasonic (or variants)
9	Set the device in pin 12 as output.	48	temperature (or variants)
10	Resistor	49	light (or variants)
11	IDE	50	motor (or variants)
12	Microcontroller	51	rgb
13	GND	52	reset
14	setup() & loop()	53	gnd (or variants)
15	OFF	54	5v (or variants)
16	1 kΩ ±5%	55	transistor
17	470 kΩ ±5%	56	LED
18	1 MΩ ±5%	57	Serial.print()
19	68 Ω ±5%	58	0 to 1023
20	39 kΩ ±5%	59	HIGH or LOW
21	8.2 Ω ±5%	60	9600
22	4.7 MΩ ±10%	61	runs repeatedly
23	47 Ω ±5%	62	LOW
24	3 kΩ ±10%	63	ohm
25	560 kΩ ±10%	64	5
26	200 Ω	65	pinMode()
27	500 Ω	66	arduino uno (or variants)
28	100 Ω	67	ldr (or variants)
29	300 Ω	68	lcd (or variants)
30	50 Ω	69	buzzer (or variants)
31	150 Ω	70	breadboard (or variants)
32	2 kΩ	71	led (or variants)
33	1.5 kΩ	72	dc motor (or variants)
34	400 Ω	73	resistor (or variants)
35	250 Ω	74	pushbutton (or variants)
36	arduino	75	servo (or variants)
37	microcontroller	76	Rubric: A working sketch that sets pinMode() for t...
38	breadboard	77	Rubric: A correct circuit: Arduino wired to a brea...
39	jumper wires (or variants)		

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